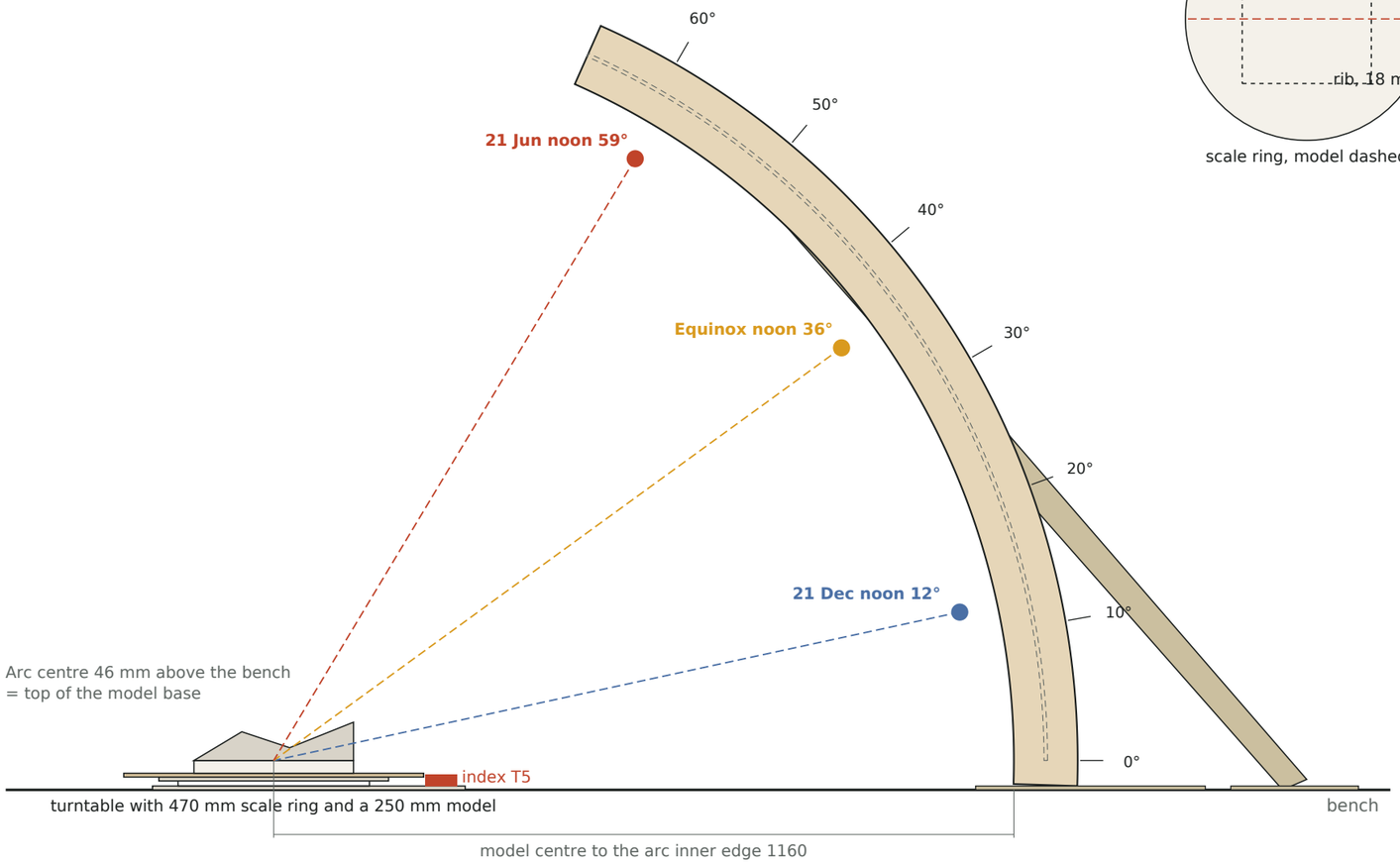
Gleniff heliodon

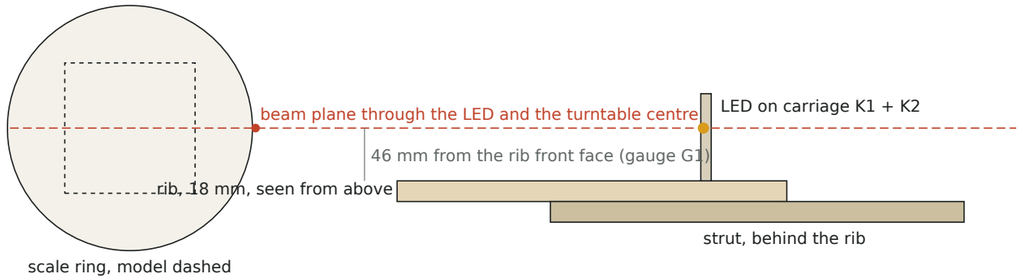
A bench sun simulator for 250 mm site models at 54.4° N. Laser-cut 6 mm plywood, one LED.

Side elevation, about 1:10 (strut behind the rib)

LED 1100 mm from the model centre. Arc band 1160–1260 mm, 3 layers of 6 mm ply (18 mm), scale 0°–64°.



Plan: lining up the beam plane (sideways ×2.2)



How it works

The model never tilts. Turn the turntable until the sun’s bearing sits at the index mark, then slide the LED along the arc to the sun’s height. Both numbers are on sheet 3, or read Sun height now and Sun bearing off the Sun & climate page for any date and time. The bearing scale is on a 470 mm ring, outside the corners of a 250 mm model, so every number stays readable. At 1:200 a 5 mm pin is a 1 m post: check its shadow against Shadow of a 1 m post on the page.

What it gets right, and what it doesn’t

The LED is 1.1 m away, so its rays spread a little: across a 250 mm model, shadows at the edges point about 6° off the true direction (9° at the corners) and are a little long. The centre is exact. A bigger model needs the projector instead. The arc reaches 64°; the highest sun at Gleniff is 59° (21 June). The carriage can’t go below about 1°.

LED

A 20–30 W COB LED, 4000–5000 K, with a 30–45° reflector or lens and a heatsink. Run the cable along the back of the rib. Dim the room for crisp shadows.

Parts and assembly

45 parts on 9 sheets of 600 × 400 mm, 6 mm plywood. Red cut, blue score, black engrave.

Part	Qty	Size mm	Sheet	What it is
AI1	1	355 × 60	7	front, inner strip -2.0°–15.0°
AI2	1	357 × 60	7	front, inner strip 15.0°–32.0°
AI3	1	357 × 60	7	front, inner strip 32.0°–49.0°
AI4	1	357 × 60	7	front, inner strip 49.0°–66.0°
AO1	1	369 × 60	6	front, outer strip, scale engraved -2.0°–15.0°
AO2	1	372 × 61	6	front, outer strip, scale engraved 15.0°–32.0°
AO3	1	372 × 61	6	front, outer strip, scale engraved 32.0°–49.0°
AO4	1	372 × 61	1	front, outer strip, scale engraved 49.0°–66.0°
BI1	1	176 × 48	4	middle, inner strip -2.0°–6.5°
BI2	1	356 × 57	8	middle, inner strip 6.5°–23.5°
BI3	1	356 × 57	8	middle, inner strip 23.5°–40.5°
BI4	1	356 × 57	8	middle, inner strip 40.5°–57.5°
BI5	1	179 × 48	4	middle, inner strip 57.5°–66.0°
BO1	1	183 × 48	4	middle, outer strip -2.0°–6.5°
BO2	1	372 × 58	6	middle, outer strip 6.5°–23.5°
BO3	1	372 × 58	6	middle, outer strip 23.5°–40.5°
BO4	1	372 × 58	6	middle, outer strip 40.5°–57.5°
BO5	1	187 × 48	3	middle, outer strip 57.5°–66.0°
C1	1	369 × 113	5	back, solid -2.0°–15.0°
C2	1	372 × 113	4	back, solid 15.0°–32.0°
C3	1	372 × 113	5	back, solid 32.0°–49.0°
C4	1	372 × 113	3	back, solid 49.0°–66.0°
SA1	1	400 × 50	9	strut, A layer, 0–400 mm from the top end
SA2	1	400 × 50	9	strut, A layer, 400–800 mm from the top end
SA3	1	429 × 50	7	strut, A layer, 800–1207 mm from the top end
SB1	1	200 × 50	3	strut, B layer, 0–200 mm from the top end
SB2	1	400 × 50	9	strut, B layer, 200–600 mm from the top end
SB3	1	400 × 50	9	strut, B layer, 600–1000 mm from the top end
SB4	1	229 × 50	4	strut, B layer, 1000–1207 mm from the top end
SC1	1	400 × 50	9	strut, C layer, 0–400 mm from the top end
SC2	1	400 × 50	9	strut, C layer, 400–800 mm from the top end
SC3	1	429 × 50	7	strut, C layer, 800–1207 mm from the top end
K1	1	152 × 90	3	carriage plate; pointer reads the altitude scale
K2	1	90 × 86	6	LED holder; tabs go into the carriage slots. Ø30 hole + M3 holes on a 30 mm …
K3	2	40 × 14	1	T-nut plates: sit in the channel under the M5 bolts (optional, instead of bolt …
T1	1	360 × 360	2	turntable disc; T3 and the scale ring are glued on top of it
T3	1	300 × 300	5	centre of the top layer; scored square shows where a 250 mm model sits
T4A	1	470 × 235	3	scale ring half, bearings 224.5°–360°–44.5°; glue on T1 around T3, joints off the …
T4B	1	470 × 235	4	scale ring half, bearings 44.5°–224.5°; glue on T1 around T3, joints off the T1 edge
T2	1	490 × 380	1	turntable base; the tab carries the index, pointing at the arc
T5	2	50 × 43	3	index pointer: glue 2 on the T2 tab, T5T on top; the tip reads the scale ring
T5T	1	50 × 43	3	index pointer, top layer: engraved line and arrow (goes on top of the two T5)
F1	1	360 × 200	2	arc foot plate; clamp to the bench
F2	1	200 × 150	6	strut foot plate
G1	1	120 × 60	5	offset gauge: rib front face to the beam plane through the turntable centre

Also needed

- 300 mm (12 in) lazy-susan bearing, with screws
- 6 mm dowel rod, about 1.5 m, cut to 18 mm lengths
- 2 × M5 × 30 bolts, washers and wing nuts (carriage)
- 2 × M5 × 50 countersunk bolts and nuts (strut)
- 4 steel angle brackets and short wood screws (feet)
- PVA wood glue, clamps, 2 G-clamps for the bench
- COB LED, heatsink, lens, driver or power supply

Assembly

1. Glue the arc in three layers. Lay the back layer C1–C4 flat, joints butted. Glue the middle strips BI1–BI5 and BO1–BO5 on top, leaving the 11 mm channel open; their joints fall halfway along the C pieces. Then glue the front strips AI1–AI4 and AO1–AO4, scale outwards, leaving the 5.5 mm slot. Push dowels through every hole as you go to keep the layers lined up. Weight it flat overnight.
2. Countersink the two strut holes on the front (they sit under the carriage’s path), bolt the strut layers SA, SB, SC together with glue, then bolt the strut to the back of the rib.
3. Stand the rib on F1 between the scored lines and fix it with two angle brackets on the back face. Fix the strut foot to F2 the same way. Clamp both plates to the bench.
4. Carriage: slide two M5 bolt heads (or K3 plates) into the channel from the top of the arc. Push K2’s tabs into K1’s slots and glue. Fit K1 over the bolts with washers and wing nuts. Mount the LED in K2 and aim it along the arc’s radius.
5. Turntable: glue T3 in the middle of T1, then the ring halves T4A and T4B around it, numbers up, N over the MODEL NORTH arrow. Glue T5, T5 and T5T into the outline on T2’s tab. Screw the lazy susan between T2 and T1, centred. Stand it with the tab towards the arc, its centre on the beam plane (G1 gauge from the rib front face) and the inner-edge distance from the arc.
6. Check it: set the carriage to 0° and sight along the LED. It should look straight at the turntable centre. Pack F1 or the turntable until the arc centre is level with the top of the model base.

Setting table for Gleniff

Sun height (altitude) and bearing for each hour, Irish clock time (summer time included). Set the carriage to the height and the turntable to the bearing.

Date	06:00	07:00	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:00	Highest
21 Jan	·	·	·	2° 129°	8° 141°	12° 155°	15° 169°	16° 184°	14° 198°	10° 212°	5° 225°	·	·	·	·	·	16° 12:45 180°
21 Feb	·	·	2° 111°	10° 123°	16° 137°	21° 152°	24° 167°	25° 183°	23° 200°	19° 215°	14° 229°	6° 242°	·	·	·	·	25° 12:48 180°
21 Mar	·	3° 94°	12° 106°	20° 119°	27° 134°	32° 150°	35° 167°	36° 186°	34° 204°	29° 221°	22° 236°	15° 249°	6° 262°	·	·	·	36° 12:41 180°
21 Apr	·	5° 77°	14° 89°	22° 101°	31° 115°	38° 130°	44° 148°	47° 168°	47° 190°	44° 211°	39° 229°	31° 244°	23° 258°	14° 271°	6° 283°	·	48° 13:33 180°
21 May	4° 60°	12° 72°	21° 84°	29° 96°	38° 109°	46° 125°	52° 144°	55° 167°	55° 192°	52° 215°	46° 235°	38° 251°	29° 264°	21° 276°	12° 288°	4° 300°	56° 13:30 180°
21 Jun	6° 58°	14° 69°	22° 81°	31° 93°	40° 106°	48° 121°	54° 141°	58° 164°	59° 191°	55° 215°	49° 236°	41° 252°	33° 265°	24° 277°	15° 289°	8° 300°	59° 13:35 180°
21 Jul	3° 58°	11° 70°	19° 82°	28° 94°	37° 107°	45° 122°	51° 141°	55° 163°	56° 188°	53° 212°	47° 232°	39° 248°	31° 262°	22° 274°	14° 286°	6° 298°	56° 13:40 180°
21 Aug	·	4° 76°	13° 88°	22° 100°	30° 113°	38° 129°	44° 146°	47° 167°	47° 188°	44° 209°	39° 228°	32° 243°	24° 257°	15° 270°	6° 282°	·	48° 13:37 180°
21 Sep	·	·	5° 96°	14° 109°	22° 122°	28° 137°	33° 154°	36° 172°	36° 190°	33° 208°	28° 224°	21° 239°	13° 252°	4° 265°	·	·	36° 13:26 180°
5 Oct site visit	·	·	2° 100°	10° 113°	18° 126°	24° 141°	28° 157°	31° 174°	30° 191°	27° 208°	22° 223°	16° 237°	8° 250°	·	·	·	31° 13:22 180°
21 Oct	·	·	·	5° 117°	13° 130°	19° 144°	23° 159°	25° 175°	24° 191°	21° 207°	16° 222°	10° 235°	2° 248°	·	·	·	25° 13:18 180°
2 Nov submission	·	·	2° 119°	9° 132°	15° 146°	19° 160°	21° 176°	20° 191°	17° 206°	13° 220°	6° 233°	·	·	·	·	·	21° 12:17 180°
21 Nov	·	·	·	4° 134°	10° 147°	14° 161°	16° 175°	15° 190°	12° 204°	8° 218°	2° 230°	·	·	·	·	·	16° 12:19 180°
21 Dec	·	·	·	·	6° 145°	10° 159°	12° 173°	12° 187°	10° 200°	6° 214°	·	·	·	·	·	·	12° 12:32 180°

Top number: height above the horizon, for the arc. Bottom (grey): bearing clockwise from north, for the turntable. A dot means the sun is down. Figures are from the same calculation as the Sun & climate page, at the mill, 54.40° N 8.40° W, for 2026; other years differ by a fraction of a degree. The mountains outside the survey can hide a low sun: the heliodon, like the model, doesn’t know they’re there.